# Evaluation runs: 2pc-ab (two-pc-abstain-kragl)

generated 2026-11-08 from brain-data 614b86250 2026-11-08 11:41:25 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 0.0 | 4 | 1 h, 250 GB | 3 | 14 | brain03, 2026-11-03 |
| 2 | 36 | 13 | conforms | 0.0 | 4 | 1 h, 250 GB | 11 | 171 | brain03, 2026-11-03 |
| 3 | 238 | 67 | conforms | 0.1 | 4 | 1 h, 250 GB | 27 | 883 | brain03, 2026-11-03 |
| 4 | 1962 | 377 | conforms | 9.9 | 6 | 1 h, 250 GB | 172 | 145711 | brain03, 2026-11-03 |
| 5 | 19744 | 2291 | conforms | 406.5 | 15 | 1 h, 250 GB | 690 | 5862635 | brain03, 2026-11-03 |
| 6 | 236938 | 15205 | conforms | 1022.7 | 30 | 1 h, 250 GB | 746 | 12379136 | brain03, 2026-11-03 |
| 7 | 3314280 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

### stateless: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 0.0 | 4 | 1 h, 250 GB | 3 | 20 | brain03, 2026-11-03 |
| 2 | 36 | 13 | 6 | violation | 0.1 | 4 | 1 h, 250 GB | 12 | 283 | brain03, 2026-11-03 |
| 3 | 238 | 67 | 18 | violation | 0.5 | 4 | 1 h, 250 GB | 56 | 8825 | brain03, 2026-11-03 |
| 4 | 1962 | 377 | 72 | violation | 9.7 | 4 | 1 h, 250 GB | 148 | 159953 | brain03, 2026-11-03 |
| 5 | 19744 | 2291 | 360 | violation | 751.6 | 19 | 1 h, 250 GB | 1065 | 13419655 | brain03, 2026-11-03 |
| 6 | 236938 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-03 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 8 off-by-one bag.t16.p.r3, 8 correct bag.t16.p.r3, 9 off-by-one bag, 9 correct bag, 9 off-by-one bag.t16.p.r3, 9 correct bag.t16.p.r3, 10 correct bag, 10 off-by-one bag.

skipped: a smaller size had reached its limit: 8 correct bag, 8 off-by-one bag, 9 off-by-one bag, 9 correct bag, 10 correct bag, 10 off-by-one bag, 11 off-by-one bag, 11 correct bag.

left out by the table rules (invalid: foreign load): 7 off-by-one bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 0.0 | 2 | 48 h, 250 GB | brain03, 2026-11-06 |
| 2 | 36 | 13 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 238 | 67 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 1962 | 377 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 5 | 19744 | 2291 | conforms | 1.5 | 4 | 1 h, 250 GB | brain02, 2026-11-03 |
| 6 | 236938 | 15205 | conforms | 8.0 | 44 | 1 h, 250 GB | brain02, 2026-11-03 |
| 7 | 3314280 | 111539 | conforms | 63.3 | 238 | 1 h, 250 GB | brain02, 2026-11-03 |
| 8 | 52994722 | 909809 | conforms | 1111.8 | 873 | 48 h, 250 GB | brain04, 2026-11-05 |
| 9 | 953561800 | 8247331 | conforms | 12426.0 | 3020 | 48 h, 250 GB | brain04, 2026-11-05 |
| 10 | | | (48 h) | | | 48 h, 250 GB | brain04, 2026-11-03 |

### stateful: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 2 | 36 | 13 | 6 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 3 | 238 | 67 | 18 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 4 | 1962 | 377 | 72 | violation | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-03 |
| 5 | 19744 | 2291 | 360 | violation | 1.2 | 8 | 1 h, 250 GB | brain02, 2026-11-03 |
| 6 | 236938 | 15205 | 2160 | violation | 7.1 | 42 | 1 h, 250 GB | brain02, 2026-11-03 |
| 7 | 3314280 | 111539 | 15120 | violation | 55.7 | 242 | 1 h, 250 GB | brain02, 2026-11-03 |
| 8 | 52994722 | 909809 | 120960 | violation | 457.1 | 838 | 48 h, 250 GB | brain04, 2026-11-05 |
| 9 | 953561800 | 8247331 | 1088640 | violation | 11247.0 | 3025 | 48 h, 250 GB | brain04, 2026-11-05 |
| 10 | | | | (48 h) | | | 48 h, 250 GB | brain04, 2026-11-03 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 8 correct bag.t16.r31, 8 correct bag.t16.r32, 8 correct bag.t16.r33, 8 correct bag.t16.r34, 8 correct bag.t16.r35, 8 correct bag.t16.r41, 8 correct bag.t16.r42, 8 correct bag.t16.r43, 8 correct bag.t16.r44, 8 correct bag.t16.r45, 11 correct bag, 11 off-by-one bag, 12 off-by-one bag, 13 off-by-one bag, 13 correct bag, 14 correct bag, 14 off-by-one bag, 15 correct bag, 15 off-by-one bag, 16 correct bag, 16 off-by-one bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.t16.c48h.r7, 8 off-by-one bag, 8 correct bag.

left out by the table rules (timeout): 9 correct bag, 9 off-by-one bag.

skipped: a smaller size had reached its limit: 10 off-by-one bag, 10 correct bag, 11 off-by-one bag, 11 correct bag.

left out by the table rules (invalid: foreign load): 11 off-by-one bag, 12 correct bag.

## offline

### offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 0.1 | 8 | 48 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain03, 2026-11-06 |
| 2 | 36 | 13 | conforms | 0.0 | 8 | 4 h, 250 GB | 0.0 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | conforms | 0.1 | 8 | 4 h, 250 GB | 0.1 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | conforms | 0.3 | 8 | 4 h, 250 GB | 0.3 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 5 | 19744 | 2291 | conforms | 1.4 | 8 | 4 h, 250 GB | 1.2 | 0.3 | 0.2 | merge | brain05, 2026-11-03 |
| 6 | 236938 | 15205 | conforms | 8.0 | 46 | 4 h, 250 GB | 7.1 | 1.3 | 0.9 | merge | brain05, 2026-11-03 |
| 7 | 3314280 | 111539 | conforms | 83.9 | 226 | 4 h, 250 GB | 78.2 | 4.3 | 5.7 | merge | brain05, 2026-11-03 |
| 8 | 52994722 | 909809 | conforms | 1414.7 | 1133 | 4 h, 250 GB | 1381.7 | 29.0 | 33.1 | merge | brain05, 2026-11-03 |
| 8 | 52994722 | 909809 | conforms | 1257.2 | 1051 | 48 h, 250 GB | 1231.1 | 17.3 | 26.2 | merge | brain04, 2026-11-05 |
| 9 | 953561800 | 8247331 | conforms | 11364.9 | 5887 | 48 h, 250 GB | 11234.0 | 109.3 | 130.9 | merge | brain03, 2026-11-03 |
| 9 | 953561800 | 8247331 | conforms | 12702.9 | 5811 | 48 h, 250 GB | 12560.0 | 106.8 | 142.9 | merge | brain04, 2026-11-06 |

### offline: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 0.2 | 8 | 4 h, 250 GB | 0.2 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 2 | 36 | 13 | 6 | violation | 0.1 | 8 | 4 h, 250 GB | 0.0 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | 9 | violation | 0.1 | 8 | 4 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | 12 | violation | 0.3 | 8 | 4 h, 250 GB | 0.3 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 5 | 19744 | 2291 | 15 | violation | 1.4 | 10 | 4 h, 250 GB | 1.2 | 0.4 | 0.2 | merge | brain05, 2026-11-03 |
| 6 | 236938 | 15205 | 18 | violation | 8.0 | 44 | 4 h, 250 GB | 7.1 | 1.0 | 0.9 | merge | brain05, 2026-11-03 |
| 7 | 3314280 | 111539 | 21 | violation | 83.8 | 232 | 4 h, 250 GB | 78.2 | 4.5 | 5.5 | merge | brain05, 2026-11-03 |
| 8 | 52994722 | 909809 | 24 | violation | 1415.4 | 1135 | 4 h, 250 GB | 1381.7 | 26.6 | 33.8 | merge | brain05, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 52994722 | 909809 | 24 | violation | 1256.2 | 1044 | 48 h, 250 GB | 1231.1 | 12.6 | 25.1 | merge | brain04, 2026-11-05 |
| 9 | 953561800 | 8247331 | 27 | violation | 11366.2 | 5884 | 48 h, 250 GB | 11234.0 | 121.4 | 132.2 | merge | brain03, 2026-11-03 |
| 9 | 953561800 | 8247331 | 27 | violation | 12702.9 | 5808 | 48 h, 250 GB | 12560.0 | 101.1 | 142.9 | merge | brain04, 2026-11-06 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 8 correct bag.r31, 8 correct bag.r32, 8 correct bag.r33, 8 correct bag.r34, 8 correct bag.r35, 8 correct bag.r41, 8 correct bag.r42, 8 correct bag.r43, 8 correct bag.r44, 8 correct bag.r45, 11 off-by-one bag, 11 correct bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.m.c48h.r7, 9 correct bag, 9 off-by-one bag.

refused (an input missing at run time): 10 correct bag, 10 off-by-one bag.

skipped: a smaller size had reached its limit: 11 off-by-one bag, 11 correct bag.

# cross offline, Java implementation against the Rust specification

## cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 2.7 | 153 | 48 h, 250 GB | 0.0 | 2.7 | 0.0 | merge | brain03, 2026-11-06 |
| 2 | 36 | 13 | conforms | 6.2 | 154 | 4 h, 250 GB | 0.0 | 6.2 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | conforms | 6.7 | 154 | 4 h, 250 GB | 0.1 | 6.7 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | conforms | 4.3 | 170 | 10 min, 16 GB | 0.2 | 4.2 | 0.1 | in-memory | brain01, 2026-11-02 |
| 5 | 19744 | 2291 | conforms | 7.9 | 492 | 10 min, 16 GB | 1.0 | 7.7 | 0.1 | in-memory | brain01, 2026-11-02 |

## cross offline, Java implementation against the Rust specification: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 6.0 | 160 | 4 h, 250 GB | 0.2 | 6.0 | 0.0 | merge | brain05, 2026-11-03 |
| 2 | 36 | 13 | 6 | violation | 6.4 | 160 | 4 h, 250 GB | 0.0 | 6.4 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | 9 | violation | 6.9 | 167 | 4 h, 250 GB | 0.1 | 6.9 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | 12 | violation | 4.2 | 174 | 10 min, 16 GB | 0.2 | 4.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 5 | 19744 | 2291 | 15 | violation | 7.9 | 473 | 10 min, 16 GB | 1.0 | 7.7 | 0.2 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** planned but never scheduled (grid points beyond the ladder): 6 off-by-one bag, 6 correct bag, 8 correct bag, 8 off-by-one bag, 9 off-by-one bag, 9 correct bag, 10 correct bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.tfspec-jmcimpl.m.c48h.r7.

skipped: a smaller size had reached its limit: 10 off-by-one bag.

# cross offline, Rust implementation against the Java specification

## cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 6.2 | 170 | 4 h, 250 GB | 6.2 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 2 | 36 | 13 | conforms | 6.7 | 154 | 4 h, 250 GB | 6.7 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | conforms | 8.4 | 159 | 4 h, 250 GB | 8.4 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | conforms | 19.6 | 1932 | 4 h, 250 GB | 19.6 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 5 | 19744 | 2291 | conforms | 118.0 | 2970 | 4 h, 250 GB | 117.8 | 0.3 | 0.1 | merge | brain05, 2026-11-03 |

## cross offline, Rust implementation against the Java specification: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 6.2 | 170 | 4 h, 250 GB | 6.2 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 2 | 36 | 13 | 6 | violation | 6.7 | 154 | 4 h, 250 GB | 6.7 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 238 | 67 | 9 | violation | 8.4 | 159 | 4 h, 250 GB | 8.4 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 1962 | 377 | 12 | violation | 19.6 | 1932 | 4 h, 250 GB | 19.6 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 5 | 19744 | 2291 | 15 | violation | 75.6 | 1656 | 10 min, 16 GB | 75.4 | 0.2 | 0.1 | in-memory | brain04, 2026-11-01 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 6 correct bag, 8 correct bag, 8 off-by-one bag, 9 off-by-one bag, 9 correct bag.

refused (an input missing at run time): 10 correct bag, 10 off-by-one bag.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 2.8 | 148 | 48 h, 250 GB | | brain03, 2026-11-06 |
| 2 | 36 | 13 | conforms | 0.8 | 175 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 3 | 238 | 67 | conforms | 1.0 | 166 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 4 | 1962 | 377 | conforms | 2.2 | 287 | 1 h, 250 GB | | brain05, 2026-11-02 |
| 5 | 19744 | 2291 | conforms | 10.2 | 864 | 1 h, 250 GB | | brain05, 2026-11-02 |

### cross online, Java implementation (JMC) against the Rust specification export: pair **off-by-one**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 0.7 | 151 | 1 h, 250 GB | 74 | brain05, 2026-11-02 |
| 2 | 36 | 13 | 6 | violation | 0.7 | 146 | 1 h, 250 GB | 42 | brain05, 2026-11-02 |
| 3 | 238 | 67 | 18 | violation | 1.1 | 163 | 1 h, 250 GB | 58 | brain05, 2026-11-02 |
| 4 | 1962 | 377 | 72 | violation | 2.1 | 275 | 1 h, 250 GB | 50 | brain05, 2026-11-02 |
| 5 | 19744 | 2291 | 360 | violation | 10.8 | 947 | 1 h, 250 GB | 54 | brain05, 2026-11-02 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): 1 off-by-one bag, 1 correct bag, 2 off-by-one bag, 2 correct bag, 3 correct bag, 3 off-by-one bag, 4 off-by-one bag, 4 correct bag, 5 off-by-one bag, 5 correct bag, 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.w16.c48h.r7.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | conforms | 3.1 | 156 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 2 | 36 | 13 | conforms | 3.4 | 156 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 3 | 238 | 67 | conforms | 4.0 | 170 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 4 | 1962 | 377 | conforms | 9.3 | 753 | 1 h, 250 GB | | brain04, 2026-11-01 |
| 5 | 19744 | 2291 | conforms | 73.1 | 1733 | 1 h, 250 GB | | brain04, 2026-11-01 |

**cross online, Rust implementation (TraceForge) against the Java specification export: pair off-by-one, model bag, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 3 | 2 | violation | 3.1 | 156 | 1 h, 250 GB | 1 | brain04, 2026-11-01 |
| 2 | 36 | 13 | 6 | violation | 3.4 | 156 | 1 h, 250 GB | 2 | brain04, 2026-11-01 |
| 3 | 238 | 67 | 18 | violation | 4.0 | 170 | 1 h, 250 GB | 3 | brain04, 2026-11-01 |
| 4 | 1962 | 377 | 72 | violation | 9.3 | 753 | 1 h, 250 GB | 3 | brain04, 2026-11-01 |
| 5 | 19744 | 2291 | 360 | violation | 73.1 | 1733 | 1 h, 250 GB | 27 | brain04, 2026-11-01 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 off-by-one bag, 9 correct bag, 10 correct bag, 10 off-by-one bag.

left out by the table rules (ok): 1 off-by-one bag, 1 correct bag, 2 off-by-one bag, 2 correct bag, 3 correct bag, 3 off-by-one bag, 4 off-by-one bag, 4 correct bag, 5 off-by-one bag, 5 correct bag.